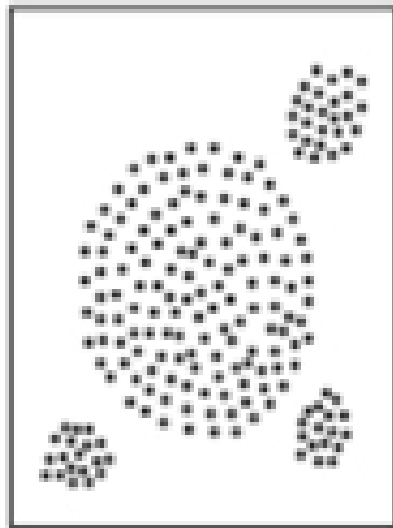


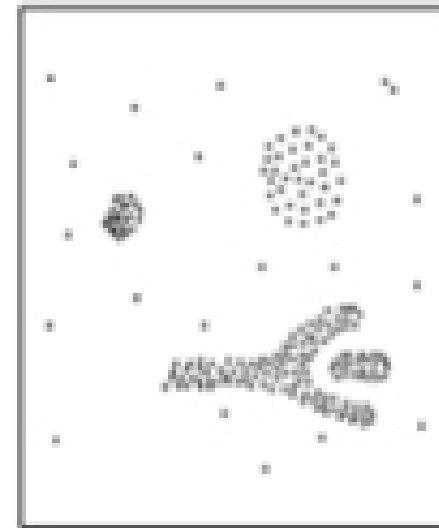
KMeans FAILS Here...



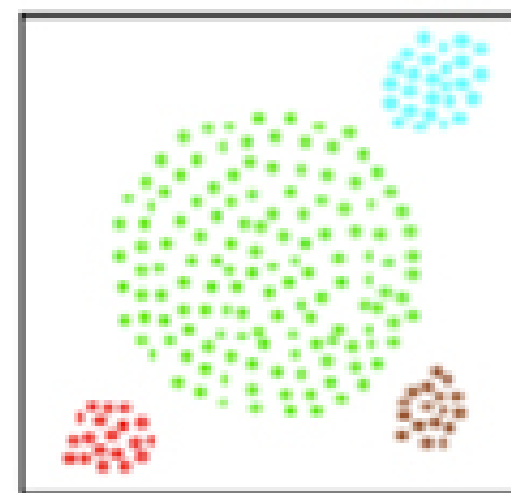
database 1



database 2



database 3



database 1



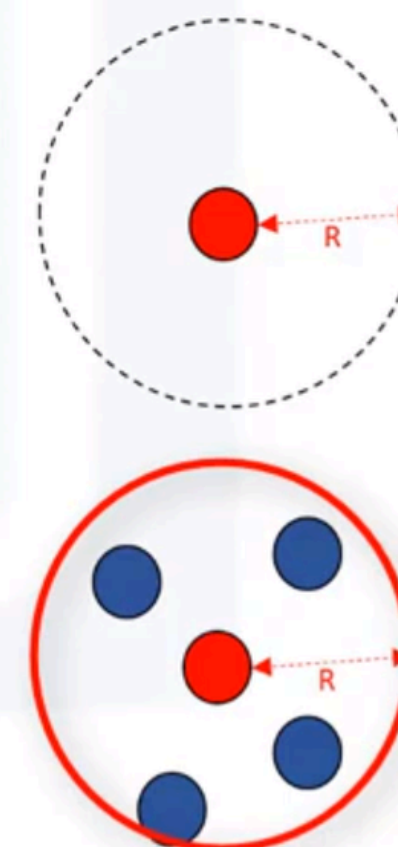
database 2



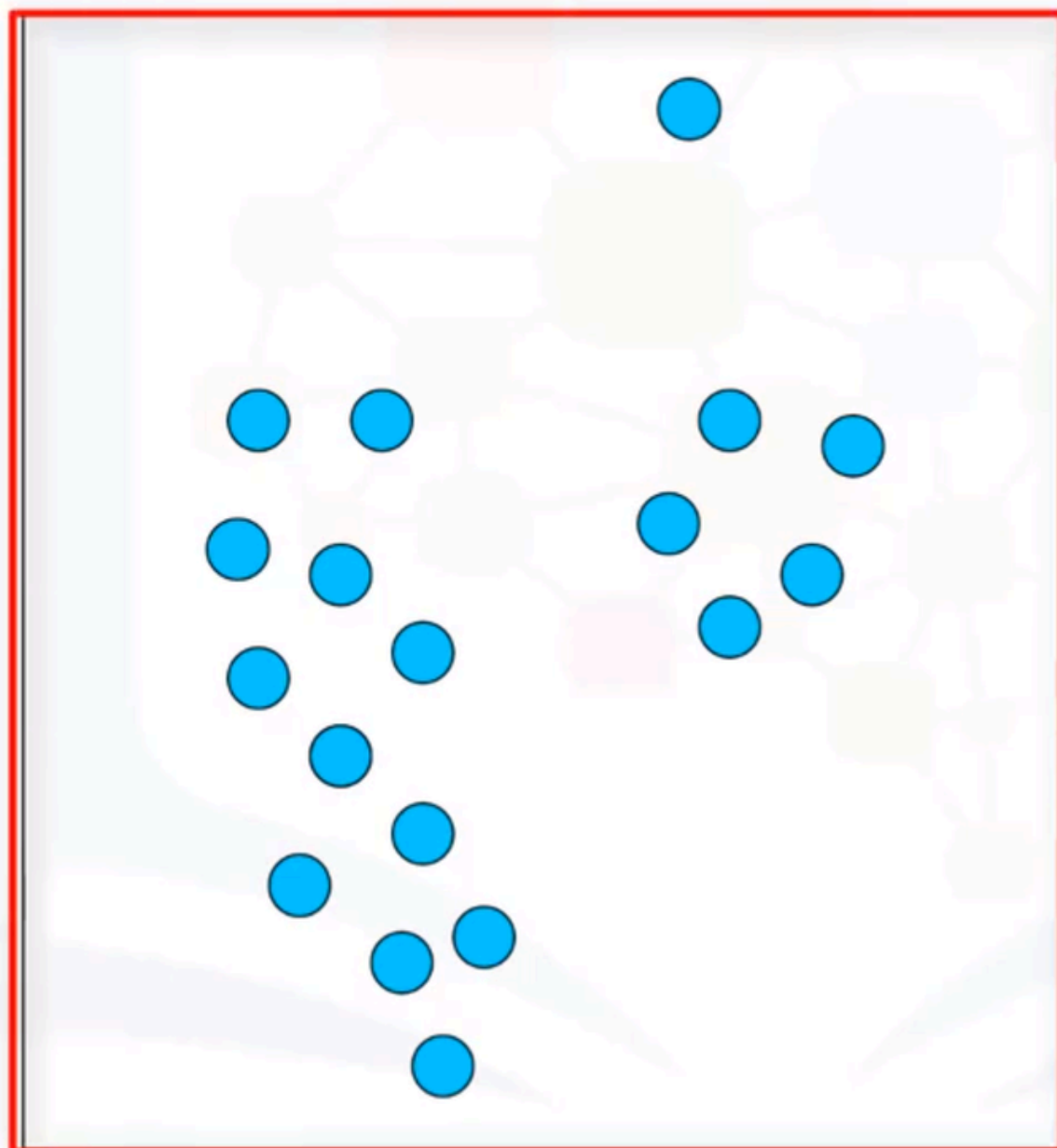
database 3

What is DBSCAN?

- DBSCAN (**D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise)
 - Is one of the most common clustering algorithms
 - Works based on density of objects
- R (**R**adius of neighborhood)
 - Radius (R) that if includes enough number of points within, we call it a dense area
- M (**M**in number of neighbors)
 - The minimum number of data points we want in a neighborhood to define a cluster



How DBSCAN works

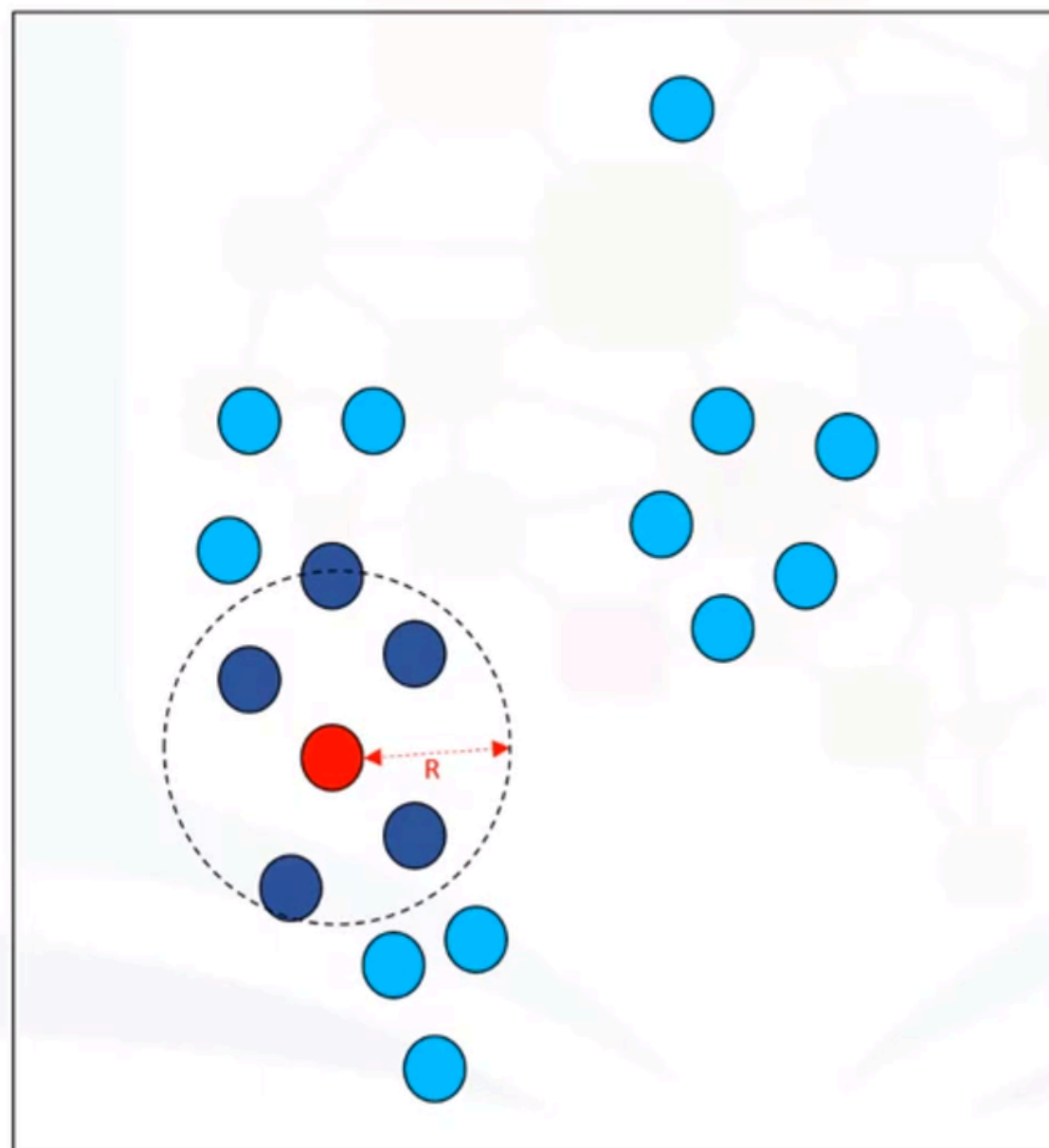


Each point is either:

- *core point*
- *border point*
- *outlier point*

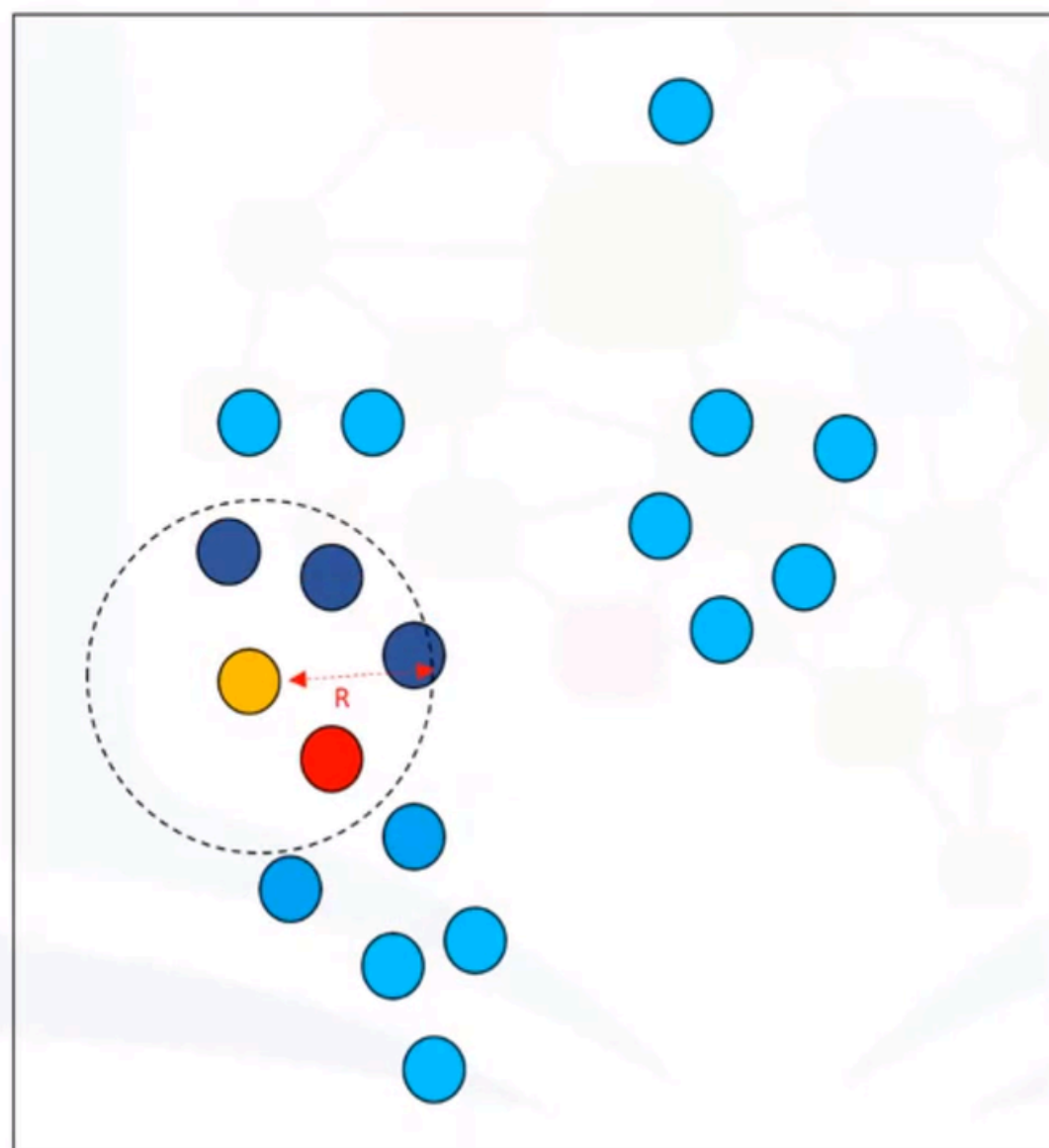
$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – core point



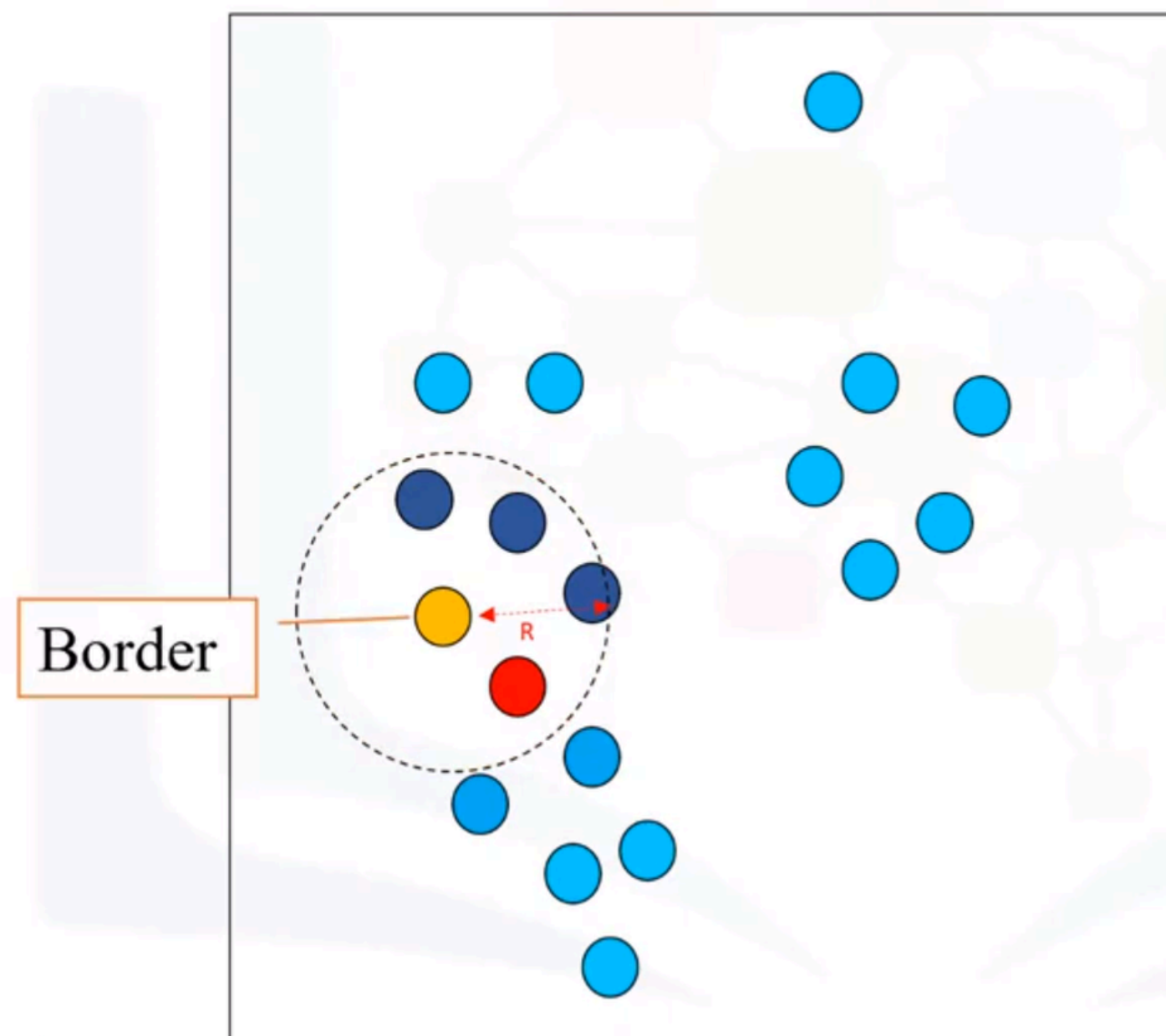
$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – border points?



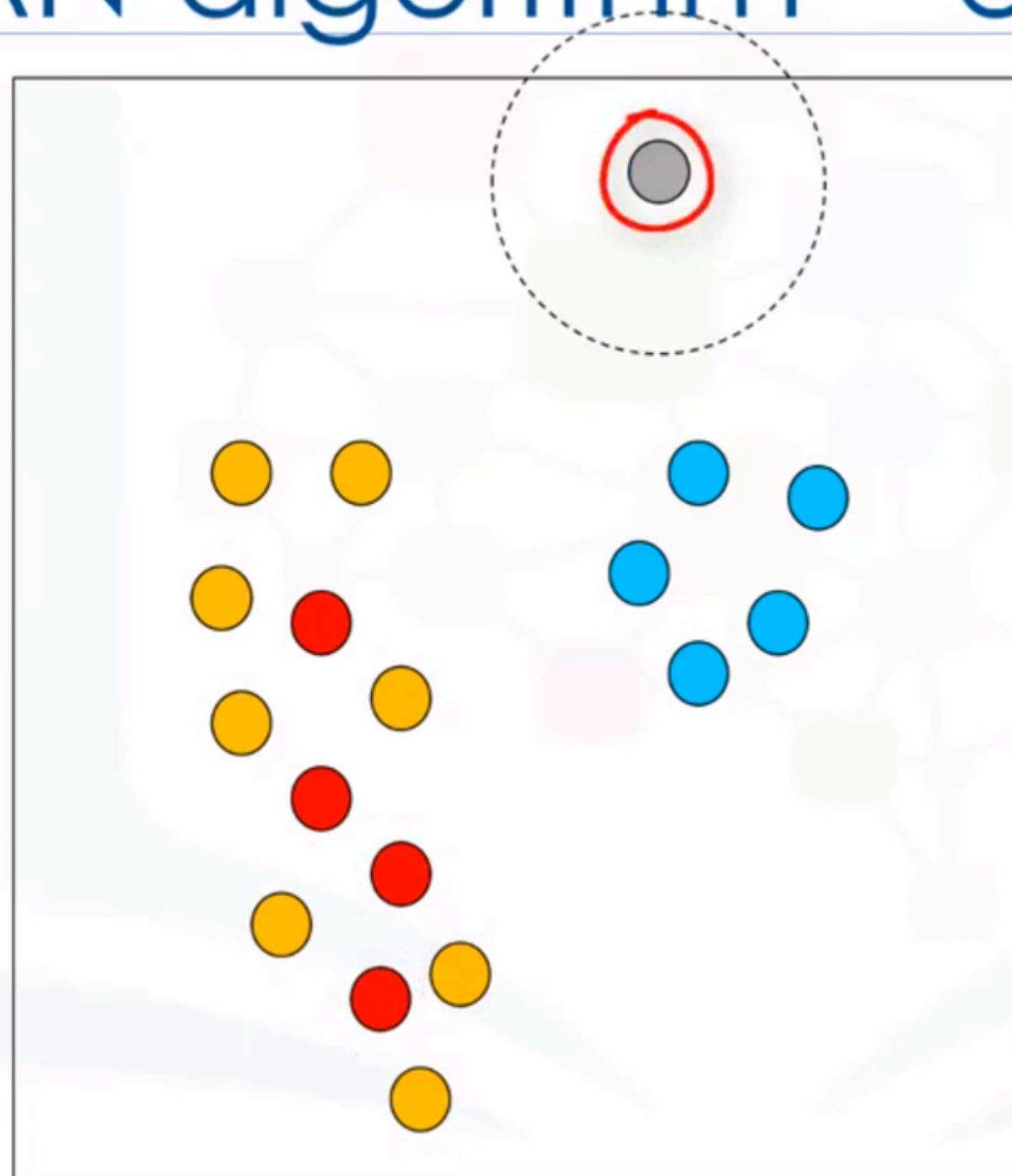
$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – border points?



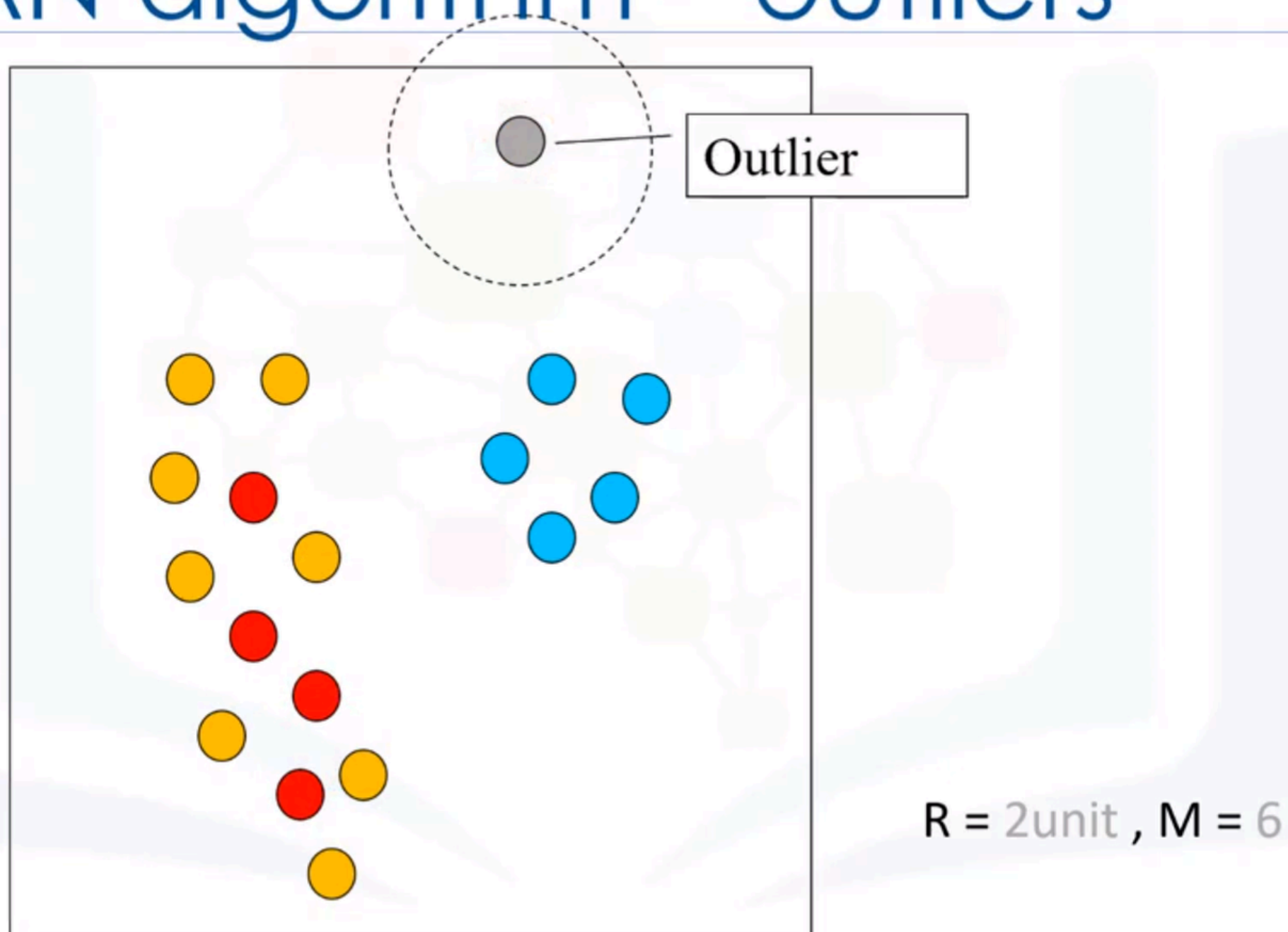
$R = 2\text{unit} , M = 6$

DBSCAN algorithm – outliers

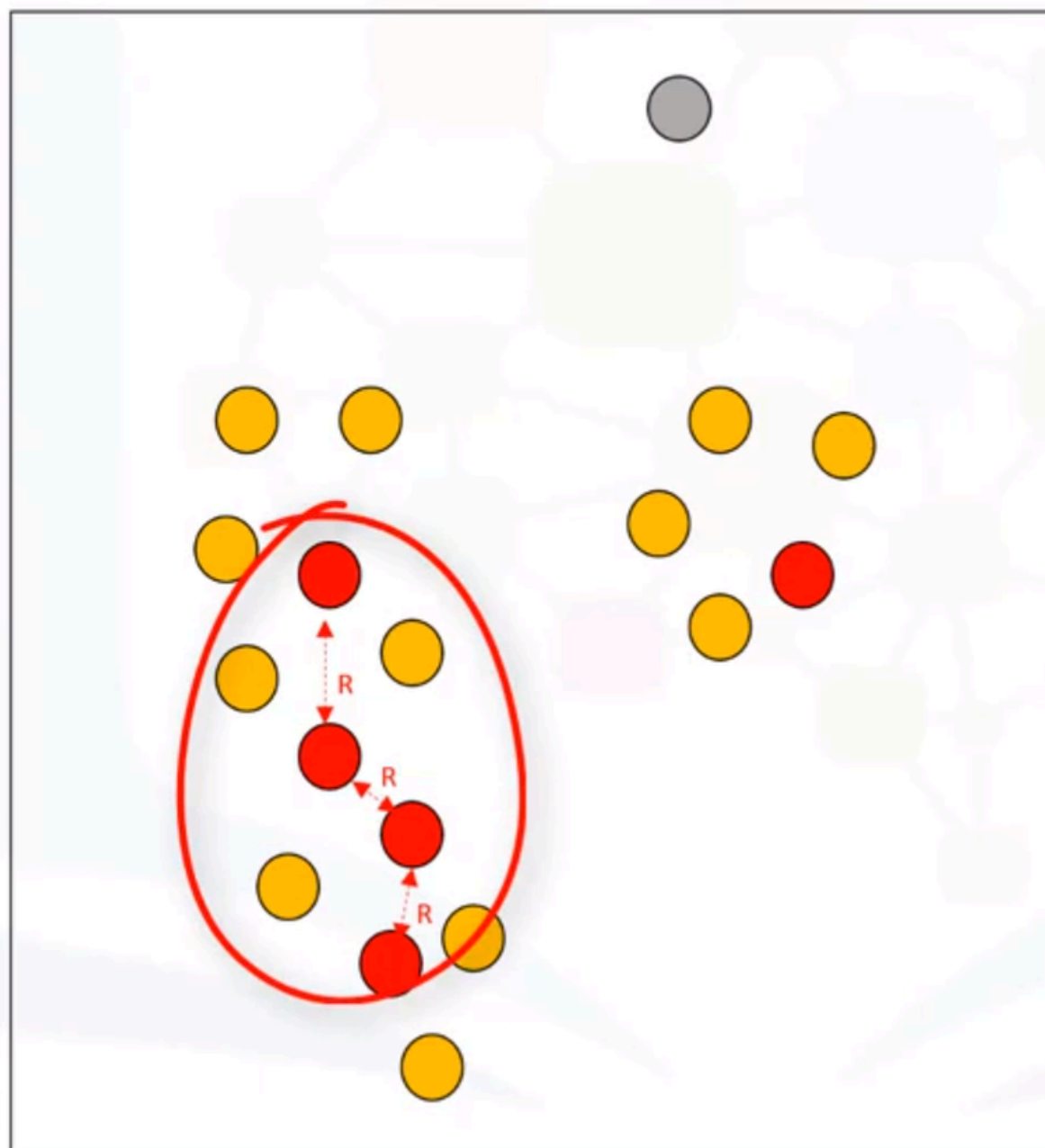


$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – outliers

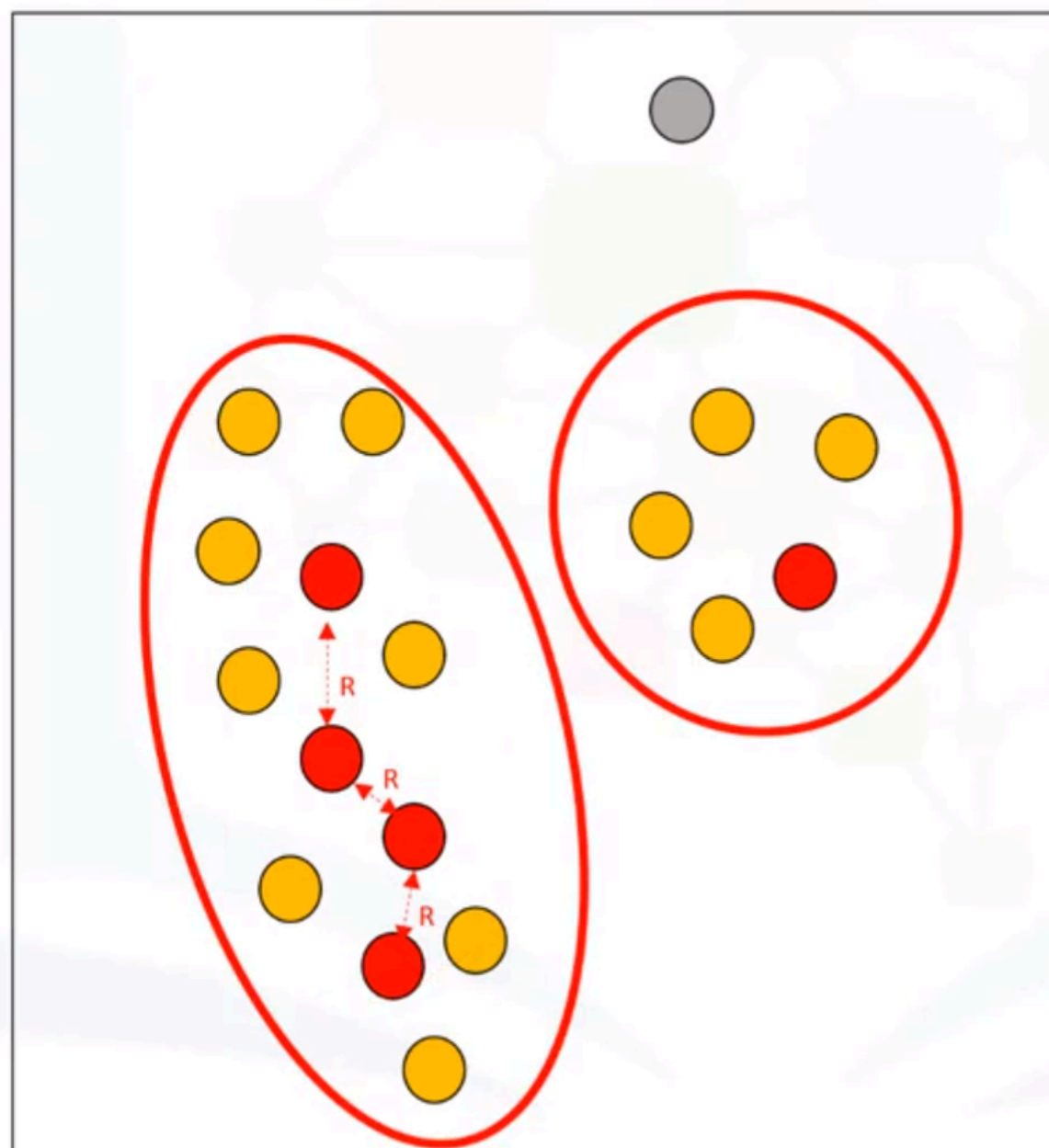


DBSCAN algorithm – clusters?



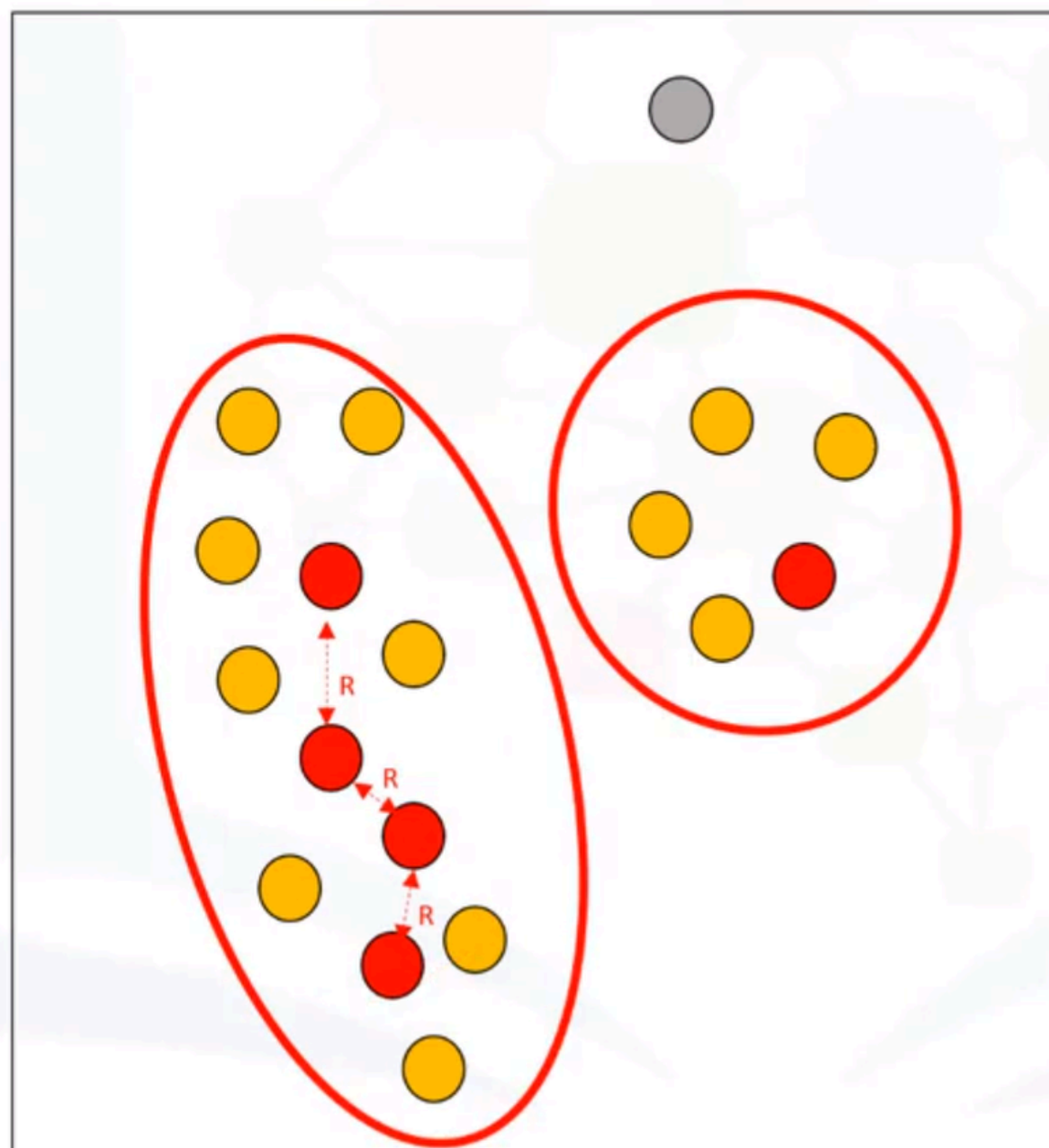
$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – clusters?



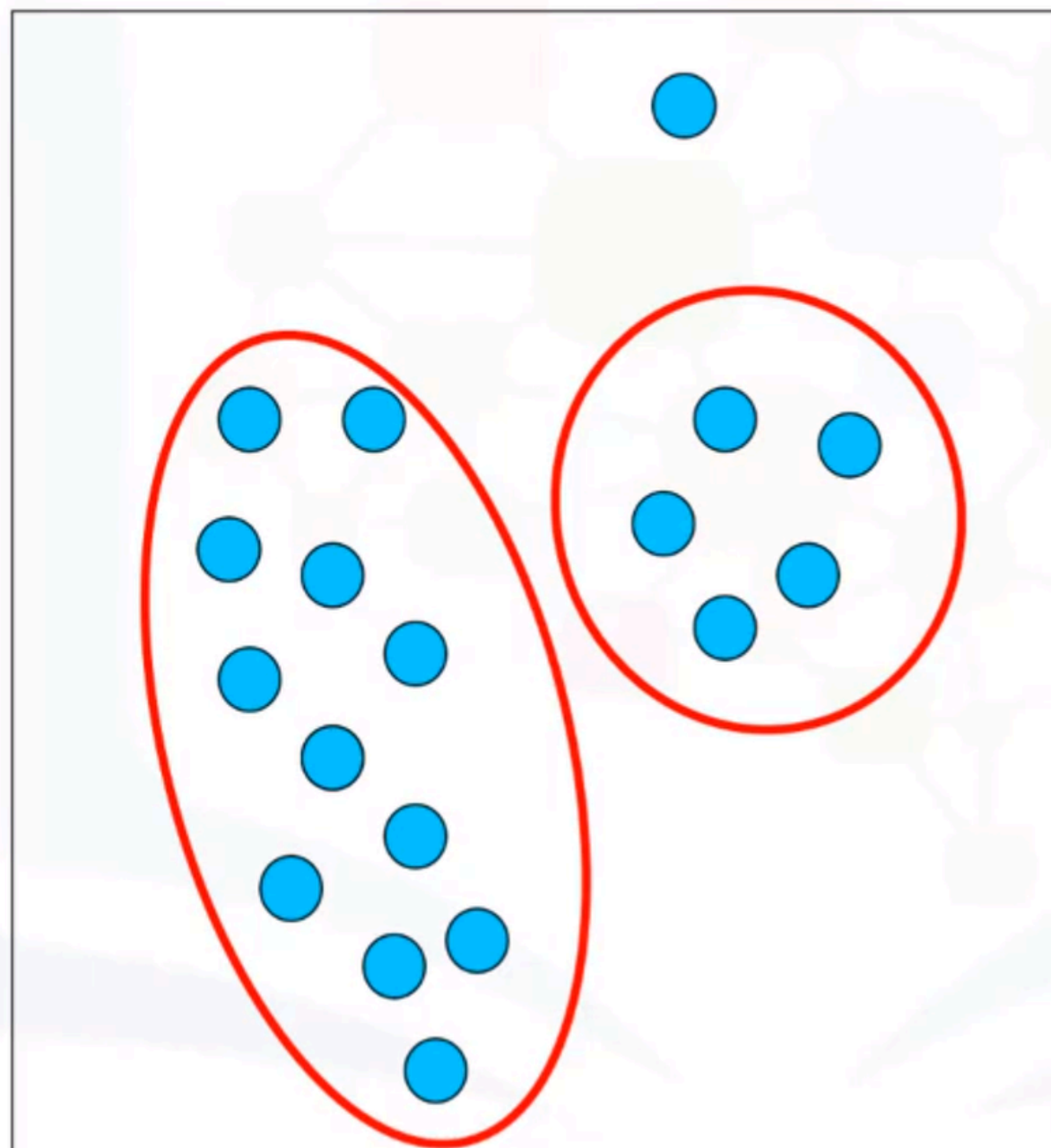
$R = 2\text{unit}$, $M = 6$

DBSCAN algorithm – clusters?



$R = 2\text{unit}$, $M = 6$

Advantages of DBSCAN



1. Arbitrarily shaped clusters
2. Robust to outliers
3. Does not require specification of the number of clusters

How DBSCAN Works?